INCH-POUND

MIL-R-52224D
7 October 1992
SUPERSEDING
MIL-R-52224C(ME)
8 January 1986

MILITARY SPECIFICATION

RETAINER ASSEMBLY, SLIDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of sliding retainer assembly for connecting a treadway-type superstructure to the universal transom.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specification and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWING

13207E9448	- Retainer Assembly, Sliding.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir, Research, Development, and Engineering Center, ATTN: SATBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951	- Standard Practice for Commercial Packaging.
D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and use of Flat Strapping Material.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A 5.1	- Carbon Steel Electrodes for Shielded Metal Arc Welding.
A 5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
D 1.1	- Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW Le Jeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The sliding retainer assembly shall be in accordance with 13207E9448 and as specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall

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provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the sliding retainer assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The sliding retainer assembly shall be fabricated from compatible materials, inherently corrosion resistant, or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the sliding retainer assembly may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the sliding retainer assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the sliding retainer assembly produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the sliding retainer assembly.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A 5.1 or A 5.18, classification E60 or E70 series, size as required.

3.5 Identification marking. The sliding retainer assembly shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

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3.6 Treatment and painting. The sliding retainer assembly shall be treated and painted in accordance with MIL-T-704, type B. Finish coat shall be green 383, color 34094, conforming to MIL-C-46168 or MIL-C-53039.

3.7 Workmanship.

3.7.1 Steel fabrication. Steel used in fabrication shall be free from kinks and sharp bends. The straightening of metal shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid over-heating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except where sharp edges or corners are required or where they are not detrimental to safety. All bends shall be made to a radius which will not cause cracking at the extreme outer fibers of the piece being bent, and all bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

3.7.2 Welders and welding.

3.7.2.1 Welders and welding operators. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.1 or the ASME code for the materials joined and the type of welding operation to be performed, and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.5).

3.7.2.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping, grinding, or wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the sliding retainer assembly to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the sliding retainer assembly shall be at the option of the contractor. Welding procedures, welding, and welds shall be in accordance with either ASME or AWS code.

3.7.3 Assembly. The sliding retainer assembly shall have no misfits or interference of mating parts that will prevent insertion of the retainer bridge pin in the various holes.

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3.7.4 Machine work. Tolerances and gauges for metal fits shall conform to the limits specified herein and on the applicable drawing and to standards of industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Parts and components. Parts and components detailed on the drawing shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawing. The drawing specifies the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.6).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article test. The first article shall be examined as specified in 4.5.1, in the order shown. Presence of one or more defects shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The sliding retainer assembly shall be examined for the following defects:

Major

101. Any part of component not in accordance with the QAP requirements as shown on the drawings (see 3.1.1).
102. Material not as specified (see 3.3).
103. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.3.1).
104. Dissimilar metals as specified in MIL-STD-889 are not effectively insulated from each other as specified (see 3.3.1.1).
105. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.1.2).
106. Components missing or not as specified (see 3.1).
107. Workmanship not as specified (see 3.7).
108. Weld defects such as slag inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal (see 3.7.2.2).
109. The bridge pin retainer cannot be inserted in the various holes. (see 3.7.3).
110. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the retainer assembly. (see 3.3.2)
111. Welders and welding operators not certified as specified (see 3.7.2.1).
112. Welding electrodes not as specified (see 3.7.2).
113. Welding and welds not as specified (see 3.7.2.2).

Minor

201. Dimensions other than those identified by QAP data not as specified. (see 3.7.4).
202. Identification marking incorrect, missing, or illegible (see 3.5).
203. Treatment and painting not as specified (see 3.6).

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105 table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

114.	Boxes not as specified for level A or B (see 5.1.1 and 5.1.2).
115.	Contents not blocked and braced in a manner to prevent free movement for level A or B (see 5.1.1 and 5.1.2).
116.	Strapping not as specified for level A (see 5.1.1).
117.	Marking missing, illegible, incorrect, or incomplete for level A, B, or Commercial (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.1.1 Level A. The sliding retainer assemblies shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitations of the applicable box specification. The contents shall be blocked and braced within the box in a manner to prevent free movement. The box shall be closed in accordance with the appendix to the applicable box specification. Strapping shall in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.1.2 Level B. The sliding retainer assemblies shall be packed together as specified in 5.1.1, except the boxes shall be domestic type or class, as applicable and strapping shall be in accordance with ASTM D 3953 and D 4675. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.3 Commercial. The sliding retainer assemblies shall be packed in accordance with ASTM D 3951.

5.2 Markings.

5.2.1 Military. In addition to any special marking specified in the contract or purchase order, marking shall be in accordance with MIL-STD-129.

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5.2.2 Commercial. Commercial markings shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sliding retainer assembly is used with the universal transom in the construction of fixed and floating bridges when the class 60 treadway-type superstructure is employed.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Quantity of sliding retainer assemblies to be furnished for first article examination, when required, and time frame required for submission of the first article (see 3.2).
d. Level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspection. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrected action taken when applicable."

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The

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applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.7.2.1	DI-MISC-80876	Welding Procedure Qualification Test Report.
3.7.2.1	DI-MISC-80875	Welding Procedures.

6.6 Definitions.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Subject term (key word) listing.

Retainer assembly
Treadway-type superstructure

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - CS

Project 5420-0212

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter shoul be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-52224D	2. DOCUMENT DATE (YYMMDD) 92/10/07

3. DOCUMENT TITLE
Retainer Assembly, Sliding

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-346 Telephone (703) 756-2340 AUTOVON 289-2340

DD FORM 1426 OCT 89 Previous editions are obsolete

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